

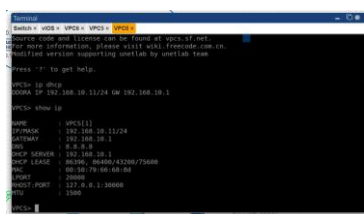
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Laporan Akhir Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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2026



3. Mengatur layanan DHCP Server pada VLAN 20 dengan menentukan alamat yang dikecualikan:
192.168.20.1 192.168.20.10

[illegible]

4. Melakukan konfigurasi alamat IP secara statis pada perangkat VPCS yang terhubung ke VLAN 30 dan VLAN 40.

```

# Switch = VBOX + VPCS + VPCD + VPCX :#
VPCS is Free Software, distributed under the terms of the "BSD" Licence.
You can use and improve can be found at vpcx-if-ai.
For more information, please visit wiki.freemove.com.cn.
I am sorry your system supporting netlib by netlib team.
press '?' to get help.

VPCSD ip 192.168.48.10/24 192.168.48.10
Checking for duplicate address...
ip: 192.168.48.10 to 255.255.255.0 gateway 192.168.48.1
VPCSD show ip
      IP:          VPCD(1)
      MAC:         32f-168-48-10/24
      VIF/MASK     32f-168-48-10/24
      GATEWAY      32f-168-48.1
      DNS
      PING         88/50.79.66.0.8s
      MTU          28460
      TX/RX PORT   10000
      RX/TX PORT   127.0.0.1:13000
      TX/RX        13000
      VPCX

```

5. Melakukan pengecekan hasil konfigurasi pada Cisco Router untuk memastikan seluruh interface dan layanan berjalan dengan baik.



The left terminal shows the configuration of a VPC named 'vpc-1' with a default route to 'vpc-1-vpccidr' and a default security group. The right terminal shows the output of the 'show ip interface brief' command, displaying the status of various interfaces including 'vpc-1-vpccidr' and 'vpc-1-vpccidr-vpc-1'.

6. Menguji konektivitas dari sisi VPCS untuk memastikan perangkat memperoleh jaringan sesuai konfigurasi.

The image displays a series of terminal commands and their outputs, organized into six distinct sections, likely representing steps in a network configuration tutorial.

- Section 1 (Top Left):** Shows the setup of hosts and interfaces.


```
Switch: vdc8 > vpc8 > vpc3 > vpc2 > vpc1 >
vpc1> show ip
NAME                < vpc3[1]
IP/PROX             192.168.18.1/24
GATEWAY             192.168.18.1
VPC SEVER           80/80, 80/80/41200/75000
VPC LEASE           00:50:79:66:68:da
PORT                20000
HOST-PORT           127.0.0.1:30000
RTU                 15000
vpc1> ping 192.168.18.1
64 bytes from 192.168.18.1: icmp_seq=1 ttl=255 time=0.348 ms
64 bytes from 192.168.18.1: icmp_seq=2 ttl=255 time=0.886 ms
64 bytes from 192.168.18.1: icmp_seq=3 ttl=255 time=0.886 ms
64 bytes from 192.168.18.1: icmp_seq=4 ttl=255 time=0.220 ms
vpc1>
```
- Section 2 (Top Right):** Shows the configuration of IP addresses and gateway on vpc1.


```
Switch: vdc8 > vpc8 > vpc3 > vpc2 > vpc1 >
vpc1> show ip
NAME                < vpc3[1]
IP/PROX             192.168.18.1/24
GATEWAY             192.168.18.1
VPC SEVER           80/80, 80/80/41200/75000
VPC LEASE           00:50:79:66:68:da
PORT                20000
HOST-PORT           127.0.0.1:30000
RTU                 15000
vpc1> ping 192.168.18.1
64 bytes from 192.168.18.1: icmp_seq=1 ttl=255 time=0.824 ms
64 bytes from 192.168.18.1: icmp_seq=2 ttl=255 time=0.825 ms
64 bytes from 192.168.18.1: icmp_seq=3 ttl=255 time=0.548 ms
vpc1>
```
- Section 3 (Middle Left):** Shows the configuration of IP addresses and gateway on vpc2.


```
Switch: vdc8 > vpc8 > vpc3 > vpc2 > vpc1 >
vpc2> show ip
NAME                < vpc3[1]
IP/PROX             192.168.18.1/24
GATEWAY             192.168.18.1
VPC SEVER           80/80, 80/80/41200/75000
VPC LEASE           00:50:79:66:68:da
PORT                20000
HOST-PORT           127.0.0.1:30000
RTU                 15000
vpc2> ping 192.168.18.1
64 bytes from 192.168.18.1: icmp_seq=1 ttl=255 time=0.899 ms
64 bytes from 192.168.18.1: icmp_seq=2 ttl=255 time=0.127 ms
64 bytes from 192.168.18.1: icmp_seq=3 ttl=255 time=0.429 ms
64 bytes from 192.168.18.1: icmp_seq=4 ttl=255 time=0.807 ms
vpc2>
```
- Section 4 (Middle Right):** Shows the configuration of IP addresses and gateway on vpc2, with a check for duplicate addresses.


```
Switch: vdc8 > vpc8 > vpc3 > vpc2 > vpc1 >
vpc2> show ip
NAME                < vpc3[1]
IP/PROX             192.168.18.1/24
GATEWAY             192.168.18.1
VPC SEVER           80/80, 80/80/41200/75000
VPC LEASE           00:50:79:66:68:da
PORT                20000
HOST-PORT           127.0.0.1:30000
RTU                 15000
vpc2> ping 192.168.18.1
64 bytes from 192.168.18.1: icmp_seq=1 ttl=255 time=0.415 ms
64 bytes from 192.168.18.1: icmp_seq=2 ttl=255 time=0.856 ms
64 bytes from 192.168.18.1: icmp_seq=3 ttl=255 time=0.927 ms
vpc2>
```
- Section 5 (Bottom Left):** Shows the configuration of IP addresses and gateway on vpc2, with a check for duplicate addresses.


```
Switch: vdc8 > vpc8 > vpc3 > vpc2 > vpc1 >
vpc2> show ip
NAME                < vpc3[1]
IP/PROX             192.168.18.1/24
GATEWAY             192.168.18.1
VPC SEVER           80/80, 80/80/41200/75000
VPC LEASE           00:50:79:66:68:da
PORT                20000
HOST-PORT           127.0.0.1:30000
RTU                 15000
vpc2> ping 192.168.18.1
64 bytes from 192.168.18.1: icmp_seq=1 ttl=255 time=0.415 ms
64 bytes from 192.168.18.1: icmp_seq=2 ttl=255 time=0.856 ms
64 bytes from 192.168.18.1: icmp_seq=3 ttl=255 time=0.927 ms
vpc2>
```
- Section 6 (Bottom Right):** Shows the configuration of IP addresses and gateway on vpc2, with a check for duplicate addresses.

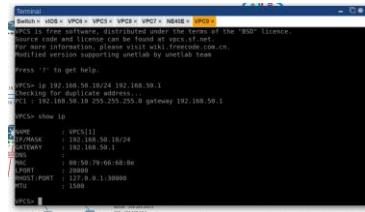

```
Switch: vdc8 > vpc8 > vpc3 > vpc2 > vpc1 >
vpc2> show ip
NAME                < vpc3[1]
IP/PROX             192.168.18.1/24
GATEWAY             192.168.18.1
VPC SEVER           80/80, 80/80/41200/75000
VPC LEASE           00:50:79:66:68:da
PORT                20000
HOST-PORT           127.0.0.1:30000
RTU                 15000
vpc2> ping 192.168.18.1
64 bytes from 192.168.18.1: icmp_seq=1 ttl=255 time=0.415 ms
64 bytes from 192.168.18.1: icmp_seq=2 ttl=255 time=0.856 ms
64 bytes from 192.168.18.1: icmp_seq=3 ttl=255 time=0.927 ms
vpc2>
```

7. Melakukan pengujian komunikasi antar VLAN dimulai dari perangkat pada VLAN 10.

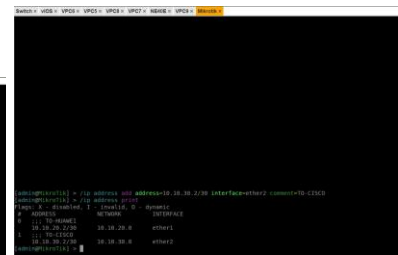
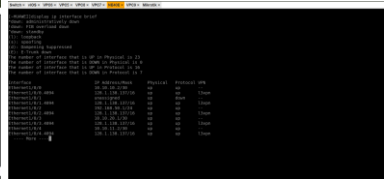
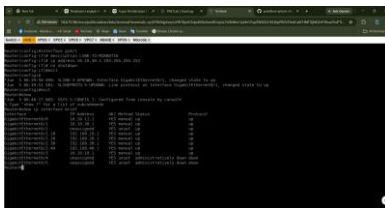
[illegible]

1.3 Konfigurasi Huawei Router dan IP Address Antarrouter

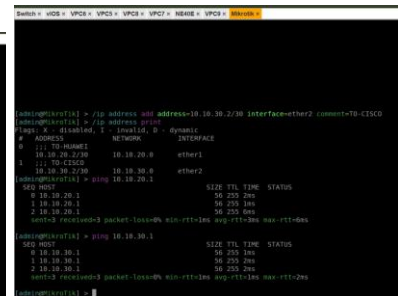
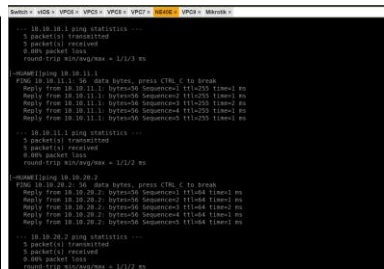
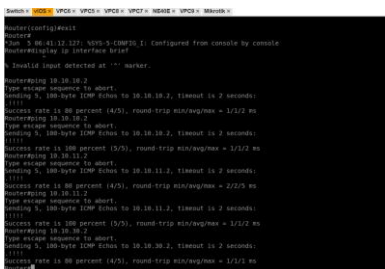
1. Mengatur alamat IP pada perangkat VPC yang berada di jaringan Huawei.



2. Melakukan konfigurasi alamat IP pada koneksi Cisco–Huawei Link 1.
3. Melakukan konfigurasi alamat IP pada koneksi Cisco–Huawei Link 2.
4. Mengkonfigurasi alamat IP untuk koneksi Huawei–MikroTik.
5. Mengatur alamat IP pada jalur komunikasi Cisco–MikroTik.
6. Melakukan verifikasi interface untuk memastikan seluruh interface aktif dan terhubung.

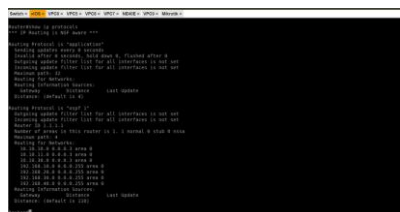


7. Melakukan pengujian konektivitas menggunakan perintah *ping* antar router.



1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Melakukan konfigurasi protokol routing OSPF pada Cisco Router.



2. Mengaktifkan OSPF pada perangkat Huawei Router.

3. Mengiklankan seluruh jaringan yang terhubung ke Huawei ke area OSPF.

4. Mengatur nilai OSPF cost pada interface Huawei untuk menentukan preferensi jalur.

5. Melakukan konfigurasi OSPF pada MikroTik serta mengiklankan seluruh network yang terhubung ke area backbone.

6. Menyesuaikan nilai OSPF cost pada interface MikroTik agar berfungsi sebagai jalur cadangan.

4

7. Memverifikasi status neighbor OSPF untuk memastikan proses adjacency berhasil.

[illegible]

8. Melakukan pengecekan routing table untuk melihat hasil pertukaran informasi routing.

[illegible]

9. Melakukan pengujian konektivitas end-to-end dari PC VLAN menuju jaringan LAN Huawei.

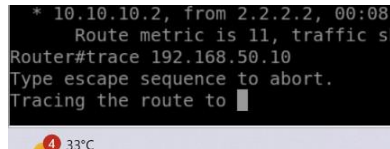
The image displays two terminal windows from a network simulation. The left window is titled 'Switch > vIOS > VPC6 > VPC5 > VPC8 > NE40E > VPC9 > Mikrotik >'. It shows the output of the 'ping' command from the switch to the PC, with successful results for all four ping attempts. The right window is titled 'Switch > vIOS > VPC6 > VPC5 > VPC8 > VPC7 > NE40E > VPC9 > Mikrotik >'. It shows the output of the 'ping' command from the switch to the PC, also with successful results for all four ping attempts. The output in both windows shows '64 bytes from 192.168.50.10' and 'icmp_seq=1' through 'icmp_seq=4' with various TTL and time values.

10. Memastikan seluruh jaringan telah dipelajari melalui OSPF dan seluruh perangkat dapat saling berkomunikasi.

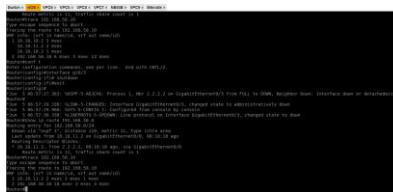
[illegible]

1.5 Pengujian Failover OSPF

1. Menguji kondisi normal ketika dua jalur Cisco–Huawei aktif dan melakukan pengecekan route menuju LAN Huawei.
2. Mensimulasikan kegagalan satu jalur dengan menonaktifkan salah satu interface menggunakan perintah *shutdown*



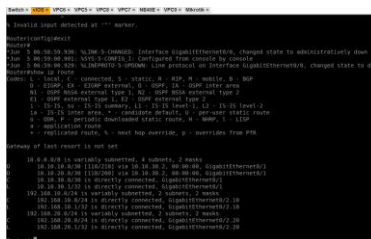
3. Melakukan pemeriksaan ulang terhadap routing table setelah perubahan jalur.



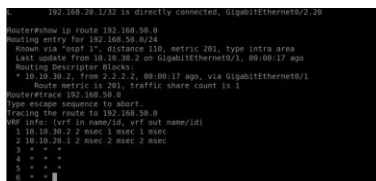
4. Menjalankan kembali proses tracing untuk memastikan trafik tetap menggunakan jalur utama.



5. Mensimulasikan kondisi kedua link Cisco–Huawei tidak aktif dengan mematikan link kedua.
6. Memeriksa kembali perubahan tabel routing setelah terjadi kegagalan total.



7. Melakukan pengujian trace untuk mengamati perpindahan trafik melalui MikroTik sebagai jalur cadangan.



8. Mengaktifkan kembali kedua link Cisco–Huawei.
9. Melakukan verifikasi akhir terhadap routing untuk memastikan jalur utama kembali digunakan.

2 Analisis Hasil Percobaan

2.1 Konfigurasi Cisco Switch

Pada praktikum pertama dilakukan konfigurasi VLAN, access port, dan trunk port pada Cisco Switch. Berdasarkan hasil verifikasi menggunakan perintah *show vlan brief*, VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing) berhasil dibuat dan berada dalam kondisi aktif. Port yang digunakan juga telah dikonfigurasi sesuai VLAN masing-masing sehingga segmentasi jaringan dapat diterapkan dengan baik. Hasil pengujian menggunakan perintah *show interfaces trunk* menunjukkan bahwa trunk berhasil membawa VLAN 10, 20, 30, dan 40 menuju Cisco Router. Kondisi ini sesuai dengan konsep trunking IEEE 802.1Q yang memungkinkan beberapa VLAN menggunakan satu jalur fisik melalui mekanisme *tagging*. Dari hasil pengujian konektivitas, perangkat dalam VLAN yang sama dapat saling berkomunikasi dengan baik, sedangkan komunikasi antar VLAN belum dapat dilakukan karena belum tersedia mekanisme *inter-VLAN routing*. Hal ini sesuai dengan teori bahwa VLAN secara default memisahkan *broadcast domain* sehingga diperlukan perangkat Layer 3 untuk menghubungkan jaringan yang berbeda. Secara keseluruhan, tujuan praktikum telah tercapai karena konfigurasi VLAN dan segmentasi jaringan berhasil diterapkan sesuai rancangan.

2.2 Konfigurasi Cisco Router

Pada praktikum kedua dilakukan penerapan metode *Router-on-a-Stick* (ROAS) menggunakan Cisco Router yang berfungsi sebagai gateway untuk seluruh VLAN sekaligus sebagai DHCP Server pada VLAN 10 dan VLAN 20. Berdasarkan hasil verifikasi, seluruh subinterface berhasil dikonfigurasi dengan alamat IP yang sesuai dan digunakan sebagai gateway pada masing-masing VLAN. Client yang berada pada VLAN 10 dan VLAN 20 berhasil memperoleh alamat IP secara otomatis melalui layanan DHCP, yang menunjukkan bahwa router mampu mendistribusikan konfigurasi jaringan kepada perangkat dengan baik. Sementara itu, perangkat pada VLAN 30 dan VLAN 40 menggunakan konfigurasi alamat IP secara statis sesuai dengan ketentuan praktikum. Hasil pengujian konektivitas menunjukkan bahwa seluruh client dapat melakukan komunikasi dengan gateway masing-masing, serta komunikasi antar VLAN juga berhasil dilakukan. Kondisi ini sesuai dengan konsep *inter-VLAN routing*, di mana router sebagai perangkat Layer 3 bertugas meneruskan paket data antar jaringan yang berbeda. Secara keseluruhan, implementasi ROAS pada praktikum ini berhasil diterapkan sesuai rancangan sehingga komunikasi antar VLAN dapat berjalan dengan baik melalui Cisco Router.

2.3 Konfigurasi Huawei Router dan IP Address Antar Router

Pada praktikum ketiga dilakukan konfigurasi alamat IP pada Huawei Router, Cisco Router, dan MikroTik Router untuk membangun konektivitas antar perangkat jaringan. Setiap koneksi *point-to-point* menggunakan subnet /30 agar alokasi alamat IP menjadi lebih efisien sesuai dengan konsep pengalamatan jaringan. Berdasarkan hasil pengujian, seluruh interface berhasil berada pada kondisi aktif (*up*) dan masing-masing router dapat melakukan komunikasi langsung melalui pengujian *ping* ke router yang terhubung. Selain itu, perangkat pada jaringan LAN Huawei juga berhasil berkomunikasi dengan gateway pada Huawei Router menggunakan jaringan 192.168.50.0/24. Keberhasilan pengujian tersebut menunjukkan bahwa konfigurasi alamat IP dan aktivasi interface telah dilakukan dengan benar. Hasil ini sesuai dengan teori bahwa komunikasi jaringan dapat berlangsung apabila perangkat berada pada subnet yang sesuai serta memiliki

koneksi fisik yang aktif. Beberapa kendala yang dapat menyebabkan kegagalan konfigurasi antara lain kesalahan penentuan subnet mask, penggunaan alamat IP yang tidak sesuai dengan tabel addressing, atau interface yang belum diaktifkan menggunakan perintah *no shutdown* maupun *undo shutdown*. Secara keseluruhan, konfigurasi jaringan pada tahap ini berhasil dilakukan sehingga konektivitas antar router dapat berjalan sesuai dengan rancangan praktikum.

2.4 Konfigurasi OSPF Multivendor dan OSPF Cost

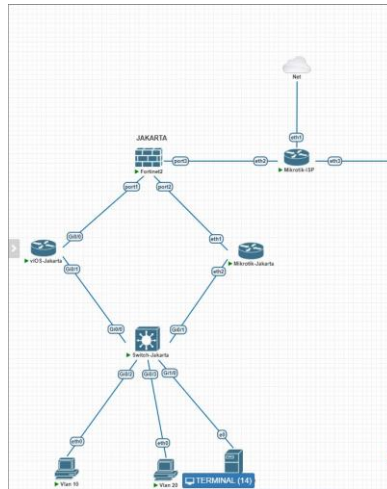
Pada praktikum keempat dilakukan implementasi protokol routing dinamis OSPF pada perangkat Cisco, Huawei, dan MikroTik dengan tujuan membangun pertukaran informasi routing secara otomatis antar perangkat dari vendor yang berbeda. Berdasarkan hasil verifikasi, seluruh router berhasil membentuk hubungan *neighbour* OSPF dengan baik. Cisco Router berhasil mengenali Huawei dan MikroTik sebagai *neighbour*, Huawei Router berhasil membangun koneksi dengan Cisco dan MikroTik, serta MikroTik juga mampu mendeteksi kedua router tersebut. Kondisi ini menunjukkan bahwa proses pertukaran *Hello Packet* dan pembentukan *adjacency* OSPF telah berjalan sesuai mekanisme protokol. Selain itu, tabel routing pada masing-masing perangkat berhasil memperoleh informasi rute secara dinamis dari jaringan lain, di mana Cisco memperoleh rute menuju jaringan LAN Huawei, Huawei memperoleh rute menuju VLAN pada Cisco, dan MikroTik menerima informasi jaringan dari kedua router lainnya. Hasil pengujian *end-to-end* menunjukkan bahwa seluruh PC pada VLAN sisi Cisco dapat berkomunikasi dengan LAN Huawei melalui *ping* dan sebaliknya. Hasil tersebut sesuai dengan teori OSPF yang menggunakan algoritma *Shortest Path First* (SPF) untuk menentukan jalur terbaik berdasarkan nilai *cost* pada setiap interface. Beberapa faktor yang dapat menyebabkan kegagalan implementasi antara lain ketidaksesuaian area OSPF, kesalahan *network advertisement*, penggunaan *router ID* yang tidak unik, maupun konfigurasi *cost* yang tidak sesuai. Secara keseluruhan, implementasi OSPF pada lingkungan multivendor berhasil dilakukan sehingga seluruh jaringan dapat saling terhubung melalui mekanisme routing dinamis.

2.5 Pengujian Failover OSPF

Pada praktikum kelima dilakukan pengujian mekanisme *failover* OSPF untuk mengetahui kemampuan jaringan dalam mempertahankan konektivitas saat terjadi gangguan pada jalur utama. Pengujian dilakukan pada kondisi seluruh link aktif, satu link utama mati, dan kedua link utama mati. Pada kondisi normal, trafik melewati jalur Cisco–Huawei karena memiliki nilai *cost* paling rendah. Ketika salah satu link dimatikan, konektivitas tetap berjalan melalui jalur yang masih aktif sehingga menunjukkan adanya redundansi jaringan. Saat kedua link utama mati, OSPF secara otomatis menghitung ulang rute dan mengalihkan trafik melalui jalur cadangan yang melewati MikroTik. Setelah link utama diaktifkan kembali, tabel routing kembali memilih jalur Cisco–Huawei sesuai nilai *cost* yang lebih rendah. Hasil pengujian menunjukkan bahwa mekanisme konvergensi dan pemilihan jalur OSPF berjalan sesuai teori sehingga jaringan tetap dapat berkomunikasi meskipun terjadi gangguan pada jalur utama.

3 Hasil Tugas Modul

3.1 TUMOD 1



Gambar 1: Screenshot topologi Jakarta

```
Switch-Jakarta > Switch-Surabaya > vIOS-Jakarta > Mikrotik-Jakarta > Ubuntu-Vlan 80 > Mikrotik-ISP > Fortinet >
Vlan 10 > Vlan 20 > Mikrotik-Surabaya > Fortinet > Vlan 30 > Vlan 40 > Topology-Vlan 40 >

Press RETURN to get started.

=====
Switchable
Switchshow vlan br
=====
VLAN Name                Status Ports
-----
1    default                active  Gi1/1, Gi1/2, Gi1/3
10   FINANCE                 active  Gi0/2
20   IT                      active  Gi0/2
60   SERVER-HQ               active  Gi1/0
1002 fddi-default            act/unsup
1003 token-ring-default    act/unsup
1004 fdinet-default        act/unsup
1005 trnet-default         act/unsup
Switch#
```

Gambar 2: Screenshot show vlan brief

```
Switch-Jakarta > Switch-Surabaya > vIOS-Jakarta > Mikrotik-Jakarta > Ubuntu-Vlan 80 > Mikrotik-ISP > Fortinet >
Vlan 10 > Vlan 20 > Mikrotik-Surabaya > Fortinet > Vlan 30 > Vlan 40 > Topology-Vlan 40 >

Switchable
Switchshow vlan br
=====
VLAN Name                Status Ports
-----
1    default                active  Gi1/1, Gi1/2, Gi1/3
10   FINANCE                 active  Gi0/2
20   IT                      active  Gi0/2
60   SERVER-HQ               active  Gi1/0
1002 fddi-default            act/unsup
1003 token-ring-default    act/unsup
1004 fdinet-default        act/unsup
1005 trnet-default         act/unsup
Switchshow interfaces trunk
=====
Port      Mode      Encapsulation  Status        Native vlan
Gi0/0     on       802.1q         trunking     1
Gi0/1     on       802.1q         trunking     1

Port      Vlans allowed on trunk
Gi0/0     10,20,60
Gi0/1     10,20,60

Port      Vlans allowed and active in management domain
Gi0/0     10,20,60
Gi0/1     10,20,60

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,60
Gi0/1     10,20,60
Switch#
```

Gambar 3: Screenshot show interfaces trunk

3.2 TUMOD 2

```
Switch-Jakarta > Switch-Surabaya > IOS-Jakarta > Mikrotik-Jakarta > Ubuntu-Vlan 10 > Mikrotik-ISP > Fortinet >
Vlan 10 > Vlan 20 > Mikrotik-Surabaya > Fortinet > Vlan 30 > Vlan 40 > Tinycore-Vlan 40 >

Press RETURN to get started.

*****
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* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****

Router#enable
Router#show ip interface brief
Interface IP-Address OK? Method Status P
GigabitEthernet0/0 10.10.100.2 YES manual up
GigabitEthernet0/1 unassigned YES unset up
GigabitEthernet0/1.10 192.168.10.2 YES manual up
GigabitEthernet0/1.20 192.168.20.2 YES manual up
GigabitEthernet0/1.60 192.168.60.2 YES manual up
GigabitEthernet0/2 unassigned YES unset administratively down do
GigabitEthernet0/3 unassigned YES unset administratively down do
Router#
```

Gambar 4: Screenshot show ip interface brief.

```
Switch-Jakarta > Switch-Surabaya > IOS-Jakarta > Mikrotik-Jakarta > Ubuntu-Vlan 10 > Mikrotik-ISP > Fortinet > Vlan 10 > Vlan 20 >
Mikrotik-Surabaya > Fortinet > Vlan 30 > Vlan 40 > Tinycore-Vlan 40 >

Press RETURN to get started.

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*****

Router#enable
Router#show ip interface brief
Interface IP-Address OK? Method Status P
GigabitEthernet0/0 10.10.100.2 YES manual up
GigabitEthernet0/1 unassigned YES unset up
GigabitEthernet0/1.10 192.168.10.2 YES manual up
GigabitEthernet0/1.20 192.168.20.2 YES manual up
GigabitEthernet0/1.60 192.168.60.2 YES manual up
GigabitEthernet0/2 unassigned YES unset administratively down do
GigabitEthernet0/3 unassigned YES unset administratively down do
Router#show vrrp brief
Interface Grp Pri Time Own Pre State Master addr Group addr
G10/1.10 10 120 3531 Y Master 192.168.10.2 192.168.10.1
G10/1.20 20 90 3648 Y Backup 192.168.20.3 192.168.20.1
G10/1.60 60 120 3531 Y Master 192.168.60.2 192.168.60.1
Router#
```

Gambar 5: Screenshot show vrrp brief.

```
Switch-Jakarta > Switch-Surabaya > IOS-Jakarta > Mikrotik-Jakarta > Ubuntu-Vlan 10 > Mikrotik-ISP > Fortinet > Vlan 10 > Vlan 20 >
Mikrotik-Surabaya > Fortinet > Vlan 30 > Vlan 40 > Tinycore-Vlan 40 >

Press RETURN to get started.

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*****

Router#enable
Router#configure terminal
Router(config)#interface GigabitEthernet0/1.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.2 255.255.255.0
Router(config-subif)#ip helper-address 192.168.60.10
Router(config-subif)#vrrp 10 ip 192.168.10.1
Router(config-subif)#vrrp 10 priority 120
Router(config-subif)#end

Router#show running-config interface gi0/1.20
Building configuration...

Current configuration : 181 bytes
!
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.2 255.255.255.0
 ip helper-address 192.168.60.10
 vrrp 20 ip 192.168.20.1
 vrrp 20 priority 90
!
end

Router#show running-config interface gi0/1.60
Building configuration...

Current configuration : 149 bytes
!
interface GigabitEthernet0/1.60
 encapsulation dot1Q 60
 ip address 192.168.60.2 255.255.255.0
 vrrp 60 ip 192.168.60.1
 vrrp 60 priority 120
!
end
Router#
```

Gambar 6: Screenshot konfigurasi subinterface.

```
Router#ping 10.10.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Router#
```

Gambar 7: Screenshot ping dari Cisco Router ke FortiGate Jakarta.

3.3 TUMOD 3

```
admin@mikrotik:~$ /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 192.168.10.3/24 192.168.10.0 vlan10-finance
1 192.168.20.3/24 192.168.20.0 vlan20-it
2 192.168.60.3/24 192.168.60.0 vlan60-server
3 10.10.101.2/30 10.10.101.0 ether1
4 I 192.168.10.1/32 192.168.10.1 vrrp10
5 I 192.168.20.1/32 192.168.20.1 vrrp20
6 I 192.168.60.1/32 192.168.60.1 vrrp60
```

Gambar 8: Screenshot /ip address print.

```
admin@mikrotik:~$ /interface vrrp print
Flags: X - disabled, I - invalid, R - running, M - master, B - backup
# NAME INTF... MAC-ADDRESS VRRP PRI INTERVAL V V3...
0 B vrrp10 vlan... 00:00:5E:00:01:0A 10 90 1s 2 ipv4
1 R vrrp20 vlan... 00:00:5E:00:01:14 20 120 1s 2 ipv4
2 B vrrp60 vlan... 00:00:5E:00:01:3C 60 90 1s 2 ipv4
```

Gambar 9: Screenshot /interface vrrp print.

```
admin@mikrotik:~$ /ip dhcp-relay print
Flags: X - disabled, I - invalid
# NAME INTERFACE DHCP-SERVER LOCAL-ADDRESS
0 XI relay1 vlan10-finance 192.168.60.10 192.168.10.3
1 XI relay2 vlan20-it 192.168.60.10 192.168.20.3
```

Gambar 10: Screenshot /ip dhcp-relay print.

```
admin@mikrotik:~$ /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, F - fib, B - bgp, O - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREP-SRC GATEWAY DISTANCE
0 A S 0.0.0.0/0 10.10.101.1 1
1 ADC 10.10.101.0/30 10.10.101.2 ether1 0
2 ADC 192.168.10.0/24 192.168.10.3 vlan10-finance 0
3 ADC 192.168.20.0/24 192.168.20.3 vlan20-it 0
4 ADC 192.168.20.1/32 192.168.20.1 vrrp20 0
5 ADC 192.168.60.0/24 192.168.60.3 vlan60-server 0
```

Gambar 11: Screenshot /ip route print.

```
admin@mikrotik:~$ ping 10.10.101.1
PING: send=64 packet=10000 size=1000 ttl=64 time=0.000ms
0 10.10.101.1 56 255 ms
1 10.10.101.1 56 255 ms
2 10.10.101.1 56 255 ms
3 10.10.101.1 56 255 ms
4 10.10.101.1 56 255 ms
5 10.10.101.1 56 255 ms
sent=6 received=6 packet=10000 min=0.000ms avg=0.000ms max=0.000ms
```

Gambar 12: Screenshot ping dari Mikrotik ke FortiGate Jakarta.

3.4 TUMOD 4

```
root@kvm:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 50:44:22:01:39:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.60.10/24 brd 192.168.60.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::5244:22ff:fe01:3900/64 scope link
        valid_lft forever preferred_lft forever
root@kvm:~#
```

Gambar 13: Screenshot ip a.

```
root@kvm:~# ip route
default via 192.168.60.1 dev eth0 proto static
192.168.60.0/24 dev eth0 proto kernel scope link src 192.168.60.10
root@kvm:~#
```

Gambar 14: Screenshot ip route.

```

root@kvm:~# sudo cat /etc/dhcp/dhcpd.conf
authoritative;

default-lease-time 600;
max-lease-time 7200;

option domain-name-servers 8.8.8.8, 1.1.1.1;

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.100 192.168.10.200;
    option routers 192.168.10.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.10.255;
}

subnet 192.168.20.0 netmask 255.255.255.0 {
    range 192.168.20.100 192.168.20.200;
    option routers 192.168.20.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.20.255;
}

subnet 192.168.60.0 netmask 255.255.255.0 {
    option routers 192.168.60.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.60.255;
}
root@kvm:~# _

```

Gambar 15: Screenshot isi file /etc/dhcp/dhcpd.conf.

```

root@kvm:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=110 time=25.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=110 time=25.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=110 time=26.0 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 25.579/25.812/26.047/0.191 ms
root@kvm:~# _

```

Gambar 16: Screenshot ping 8.8.8.8.

3.5 TUMOD 5

```
Switch-Jakarta x Switch-Sura... x vIOS-Jakarta x Mikrotik-Jak... x Ubuntu-Vlan... x Mikrotik-ISP x Fortinet3 x Vlan 10 x Vlan 20 x
Mikrotik-Sur... x Fortinet3 x Vlan 30 x Vlan 40 x Tinycore-Via... x
FortiGate-VM64-KVM # Timeout

FortiGate-VM64-KVM login: admin
Password:
Welcome!

FortiGate-VM64-KVM # get system interface physical
== [onboard]
==[port1]
mode: static
ip: 10.10.100.1 255.255.255.252
ipv6: ::/0
status: up
speed: 10000Mbps (Duplex: full)
FEC: none
FEC_cap: none
==[port2]
mode: static
ip: 10.10.101.1 255.255.255.252
ipv6: ::/0
status: up
speed: 10000Mbps (Duplex: full)
FEC: none
FEC_cap: none
==[port3]
mode: static
ip: 10.0.12.2 255.255.255.252
ipv6: ::/0
status: up
speed: 10000Mbps (Duplex: full)
FEC: none
--More--
```

Gambar 17: Screenshot get system interface physical.

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
V - BGP VPNv4
* - candidate default

Routing table for VRF=0
* 0.0.0.0/0 [1/0] via 10.0.12.1, port3, [1/0]
10.0.12.0/30 is directly connected, port3
10.10.100.0/30 is directly connected, port1
10.10.101.0/30 is directly connected, port2
172.16.0.1/32 is directly connected, GRE-JKT-SBY
172.16.0.2/32 is directly connected, GRE-JKT-SBY
192.168.10.0/24 [10/0] via 10.10.100.2, port1, [1/0]
[10/0] via 10.10.101.2, port2, [1/0]
192.168.20.0/24 [10/0] via 10.10.100.2, port1, [1/0]
[10/0] via 10.10.101.2, port2, [1/0]
E2 192.168.30.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:31:10, [1/0]
E2 192.168.40.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:31:10, [1/0]
192.168.60.0/24 [10/0] via 10.10.100.2, port1, [1/0]
[10/0] via 10.10.101.2, port2, [1/0]
```

Gambar 18: Screenshot get router info routing-table all

```
Switch-Jakarta x Switch-Sura... x vIOS-Jakarta x Mikrotik-Jak... x Ubuntu-Vlan... x Mikrotik-ISP x Fortinet3 x Vlan 10 x Vlan 20 x Mikr
edit 1
set name "VLAN-TO-INTERNET"
set uuid 796c0b12-6504-51f1-b7bf-5abaead7acce
set srcintf "port2" "port1"
set dstintf "port3"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
edit 2
set name "LAN-to-GRE"
set uuid 010a523c-6509-51f1-84ce-e4c190359657
set srcintf "port1" "port2"
set dstintf "GRE-JKT-SBY"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
edit 3
set name "GRE-to-LAN"
set uuid 1992c959-6509-51f1-f5a6-51dd8fa54dba
set srcintf "GRE-JKT-SBY"
set dstintf "port1" "port2"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
```

Gambar 19: Screenshot firewall policy.

```

FortiGate-VM64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=112 time=22.9 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=22.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=22.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=23.0 ms
^C
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 22.3/22.6/23.0 ms

```

Gambar 20: Screenshot ping 8.8.8.8.

```

FortiGate-VM64-KVM # execute ping 172.16.0.2
PING 172.16.0.2 (172.16.0.2): 56 data bytes
64 bytes from 172.16.0.2: icmp_seq=0 ttl=255 time=2.9 ms
64 bytes from 172.16.0.2: icmp_seq=1 ttl=255 time=1.6 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=255 time=1.2 ms
^C
--- 172.16.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.2/1.9/2.9 ms

```

Gambar 21: Screenshot ping ke IP tunnel Surabaya.

```

FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	Full/ -	00:00:37	172.16.0.2	GRE-JKT-SBY

Gambar 22: Screenshot get router info ospf neighbor.

```

FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
0 E2   192.168.30.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]
0 E2   192.168.40.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]

```

Gambar 23: Screenshot get router info routing-table ospf

3.6 TUMOD 6

```
Switch-Jakarta x Switch-Sura... x VIOS-Jakarta x Mikrotik-Jak... x Ubuntu-Vlan... x Mikrotik-ISP x FortiGate
Mikrotik-Sur... x FortiGate x Vlan 10 x Vlan 40 x Tempore-Via... x

admin@mikrotik> /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 10.4.89.179/24 10.4.89.0 ether1
1 10.0.12.1/30 10.0.12.0 ether2
2 10.0.13.1/30 10.0.13.0 ether3
admin@mikrotik>
```

Gambar 24: Screenshot /ip address print hasil di ether 1 bisa saja berbeda karena emang dinamic.

```
admin@mikrotik> /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.89.1 1
1 ADC 10.0.12.0/30 10.0.12.1 ether2 0
2 ADC 10.0.13.0/30 10.0.13.1 ether3 0
3 ADC 10.4.89.0/24 10.4.89.179 ether1 0
```

Gambar 25: Screenshot /ip route print.

```
admin@mikrotik> /ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1
```

Gambar 26: Screenshot /ip firewall nat print.

```
admin@mikrotik> ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=64 time=1.3 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=1.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=1.1 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.0/1.1/1.3 ms
```

Gambar 27: Screenshot ping 8.8.8.8.

```
FortiGate-VM64-KVM # execute ping 10.0.13.2
PING 10.0.13.2 (10.0.13.2): 56 data bytes
64 bytes from 10.0.13.2: icmp_seq=0 ttl=254 time=1.3 ms
64 bytes from 10.0.13.2: icmp_seq=1 ttl=254 time=1.0 ms
64 bytes from 10.0.13.2: icmp_seq=2 ttl=254 time=1.1 ms
^C
--- 10.0.13.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.0/1.1/1.3 ms

FortiGate-VM64-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.5 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.1 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=0.6 ms
^C
--- 10.0.12.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/3.7/8.6 ms
```

Gambar 28: Screenshot ping antar-WAN FortiGate.

3.7 TUMOD 7

```
Switch-Jakarta > show vlan brief
Press RETURN to get started.

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Cisco in writing.
*****
Switch-Enable
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1  default                active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
10  SALES                   active    Gi1/1
100  OPERATIONS              active    Gi1/2, Gi1/3
1002 fddi-default           act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch#
```

Gambar 29: Screenshot show vlan brief.

```
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
Gi1/0/0   on        802.1q         trunking      1

Port      Vlans allowed on trunk
Gi1/0/0   30,40

Port      Vlans allowed and active in management domain
Gi1/0/0   30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi1/0/0   30,40
Switch#
```

Gambar 30: Screenshot show interfaces trunk.

```
admin@MikroTik > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 192.168.30.1/24 192.168.30.0 vlan30-sales
1 192.168.40.1/24 192.168.40.0 vlan40-operations
2 10.10.200.2/30 10.10.200.0 ether1
```

Gambar 31: Screenshot /ip address print.

```
admin@MikroTik > /ip dhcp server print
Flags: D - dynamic, X - disabled, I - invalid
# NAME INTERFACE ADDRESS-POOL LEASE-TIME ADD-ARP
0 dhcp30 vlan30-sales pool30 10m
```

Gambar 32: Screenshot /ip dhcp-server print.

```
admin@MikroTik > /ip pool print
# NAME RANGES
0 pool30 192.168.30.100-192.168.30.200
```

Gambar 33: Screenshot /ip pool print.

```
admin@MikroTik > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
I - inactive, U - unreachable, P - probabil
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 S 0.0.0.0/0 10.10.200.2 ether1 1
1 ADC 10.10.200.0/30 192.168.30.1 vlan30-sales 0
2 ADC 192.168.30.0/24 192.168.30.1 vlan40-operations 0
3 ADC 192.168.40.0/24 192.168.40.1 vlan40-operations 0
```

Gambar 34: Screenshot /ip route print.

```
Switch-Jakarta > Switch-Bura... > VIO-Jakarta > MikroTik-Jak... > Ubuntu-Vlan... > MikroTik-ISP > Fortinet2 > VIO
MikroTik-Bur... > Fortinet2 > Vlan20 > Vlan40 > Tangerang-Vlan... >

RTU : 1500

VPCS> ping 192.168.30.1
192.168.30.1 icmp_seq=1 timeout
^C
VPCS> ip dhcp
MORA IP 192.168.30.200/24 GW 192.168.30.1
VPCS> ping 8.8.8.8
64 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=25.800 ms
64 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=29.143 ms
64 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=25.801 ms
^C
VPCS> ip dhcp
MORA IP 192.168.30.200/24 GW 192.168.30.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.30.200/24
GATEWAY : 192.168.30.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.30.1
DHCP LEASE : 500, 600/200/525
MAC : 60:50:79:146:180:9d
I-PORT : 20000
PHOST-PORT : 127.0.0.1:30000
RTU : 1500
VPCS> |
```

Gambar 35: Screenshot client VLAN 30 mendapat IP DHCP.

```
VPCS> ping 8.8.8.8
64 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=25.800 ms
64 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=29.143 ms
64 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=25.801 ms
^C
```

Gambar 36: Screenshot ping client Surabaya ke 8.8.8.8.

3.8 TUMOD 8

```
FortiGate-VM64-KVM # Timeout

FortiGate-VM64-KVM login: admin
Password:
Welcome!

FortiGate-VM64-KVM # get system interface physical
== [onboard]
== [port1]
    mode: static
    ip: 10.0.13.2 255.255.255.252
    ipv6: ::0
    status: up
    speed: 10000Mbps (Duplex: full)
    FEC: none
    FEC_cap: none
== [port2]
    mode: static
    ip: 10.10.200.1 255.255.255.252
    ipv6: ::0
    status: up
    speed: 10000Mbps (Duplex: full)
    FEC: none
    FEC_cap: none
== [port3]
    mode: static
    ip: 0.0.0.0 0.0.0.0
    ipv6: ::0
    status: down
    speed: n/a
    FEC: none
--More--
```

Gambar 37: Screenshot get system interface physical

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E2 - OSPF external type 2, E1 - OSPF external type 1
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter are
       * - candidate default
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
* 0.0.0.0/0 [1/0] via 10.0.13.1, port1, [1/0]
C 10.0.13.0/24 is directly connected, port1
C 10.10.200.0/24 is directly connected, port2
C 172.16.0.1/24 is directly connected, GRE-SBY-JKT
C 172.16.0.2/24 is directly connected, GRE-SBY-JKT
E2 192.168.10.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:45:19, [1/0]
E2 192.168.20.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:45:19, [1/0]
E 192.168.30.0/24 [180/0] via 10.10.200.2, port2, [1/0]
E2 192.168.60.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:45:19, [1/0]
```

Gambar 38: screenshot get router info routing-table all.

```
config firewall policy
edit 1
set name "LAN-to-INTERNET"
set uuid 1c3ce4bc-6503-51f1-2b6b-03775fdd8f42
set srcintf "port2"
set dstintf "port1"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
edit 2
set name "LAN-to-GRE"
set uuid 447f60e8-6509-51f1-25b6-83c7a93f26af
set srcintf "port2"
set dstintf "GRE-SBY-JKT"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
edit 3
set name "GRE-to-LAN"
set uuid 448c7742-6509-51f1-a24f-2ca1b8f7fb82
set srcintf "GRE-SBY-JKT"
set dstintf "port2"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
```

Gambar 39: Screenshot firewall policy

```
FortiGate-VM64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=112 time=22.9 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=22.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=23.3 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 22.2/22.8/23.3 ms
```

Gambar 40: Screenshot ping ke 8.8.8.8

```
FortiGate-VM64-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=1.2 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=1.1 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=1.2 ms
^C
--- 172.16.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/1.1/1.2 ms
```

Gambar 41: Screenshot ping ke IP tunnel Jakarta.

```
FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID    Pri  State           Dead Time   Address      Interface
1.1.1.1        1    Full/ -         00:00:31    172.16.0.1   GRE-SBY-JKT
```

Gambar 42: Screenshot get router info ospf neighbor.

```
FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
O E2   192.168.10.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
O E2   192.168.20.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
O E2   192.168.60.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
```

Gambar 43: Screenshot get router info routing-table ospf.

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```
FortiGate-VM64-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.5 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.1 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=8.6 ms
^C
--- 10.0.12.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/3.7/8.6 ms
```

```
FortiGate-VM64-KVM # execute ping 10.0.13.2
PING 10.0.13.2 (10.0.13.2): 56 data bytes
64 bytes from 10.0.13.2: icmp_seq=0 ttl=254 time=1.3 ms
64 bytes from 10.0.13.2: icmp_seq=1 ttl=254 time=1.0 ms
64 bytes from 10.0.13.2: icmp_seq=2 ttl=254 time=1.1 ms
^C
--- 10.0.13.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.0/1.1/1.3 ms
```

Gambar 44: Screenshot ping WAN antar-FortiGate.

```
FortiGate-VM64-KVM # execute ping 172.16.0.2
PING 172.16.0.2 (172.16.0.2): 56 data bytes
64 bytes from 172.16.0.2: icmp_seq=0 ttl=255 time=3.9 ms
64 bytes from 172.16.0.2: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=255 time=1.2 ms
^C
--- 172.16.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.2/2.2/3.9 ms
```

```
FortiGate-VM64-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=4.4 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=1.4 ms
^C
--- 172.16.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.4/2.4/4.4 ms
```

Gambar 45: Screenshot ping tunnel antar-FortiGate.

```
FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID    Pri  State           Dead Time   Address      Interface
2.2.2.2        1    Full/ -         00:00:37    172.16.0.2   GRE-JKT-SBY
```

Gambar 46: Screenshot get router info ospf neighbor.

```

FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
0 E2 192.168.30.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]
0 E2 192.168.40.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]

FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
0 E2 192.168.10.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
0 E2 192.168.20.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
0 E2 192.168.60.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]

```

Gambar 47: Screenshot get router info routing-table ospf.

```

VPCS> ping 192.168.40.10

64 bytes from 192.168.40.10 icmp_seq=1 ttl=60 time=13.084 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=60 time=11.700 ms
64 bytes from 192.168.40.10 icmp_seq=3 ttl=60 time=10.312 ms
^C
VPCS>

```

Gambar 48: Screenshot ping client Jakarta ke client Surabaya.

```

VPCS> ping 192.168.10.101

64 bytes from 192.168.10.101 icmp_seq=1 ttl=60 time=12.882 ms
64 bytes from 192.168.10.101 icmp_seq=2 ttl=60 time=7.930 ms
64 bytes from 192.168.10.101 icmp_seq=3 ttl=60 time=7.014 ms
^C

```

Gambar 49: Screenshot ping client Surabaya ke client Jakarta.

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```

VPCS> ip dhcp
DDORA IP 192.168.10.102/24 GW 192.168.10.1

```

Gambar 50: Screenshot IP DHCP client Jakarta (Vlan 10).

```

VPCS> ip dhcp
DDORA IP 192.168.30.200/24 GW 192.168.30.1

```

Gambar 51: Screenshot IP DHCP client Surabaya (Vlan 20).

```

VPCS> ping 8.8.8.8

64 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=28.083 ms
64 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=25.015 ms
64 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=28.061 ms
^C
VPCS>

```

Gambar 52: Screenshot ping internet dari Jakarta.

```

VPCS> ping 8.8.8.8

64 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=25.738 ms
64 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=25.469 ms
64 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=26.359 ms
64 bytes from 8.8.8.8 icmp_seq=4 ttl=110 time=26.912 ms
^C

```

Gambar 53: Screenshot ping internet dari Surabaya.

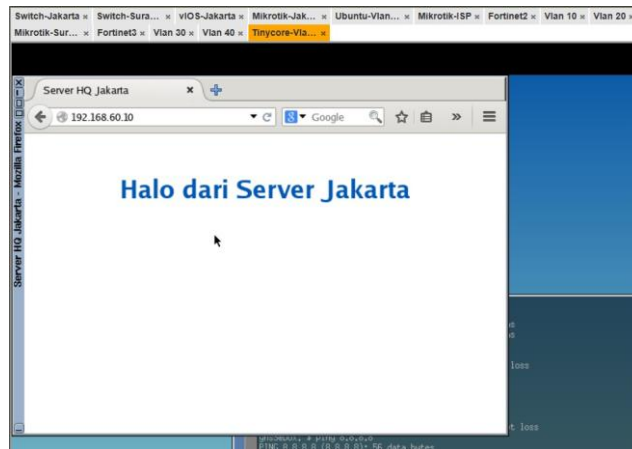
```

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=8.299 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=4.425 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.837 ms
^C
VPCS>

```

Gambar 54: Screenshot ping antar-site



Gambar 55: Screenshot akses web server Jakarta dari Surabaya.

```

FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID    Pri  State           Dead Time   Address        Interface
1.1.1.1        1   Full/-         00:00:31   172.16.0.1    GRE-SBY-JKT

```

Gambar 56: Screenshot routing table OSPF

```

VPCS> trace 192.168.40.10
Trace to 192.168.40.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.2    11.510 ms  2.193 ms  4.470 ms
 2  10.10.100.1     8.313 ms  3.927 ms  2.594 ms
 3  172.16.0.2      5.450 ms  5.588 ms  11.021 ms
 4  10.10.200.2     5.805 ms  5.475 ms  4.767 ms
 5  *192.168.40.10 11.328 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 57: Analisis singkat jalur traffic Jakarta ke Surabaya.

4 Kesimpulan

Berdasarkan hasil praktikum yang telah dilakukan, seluruh tujuan pada Modul 5 berhasil dicapai. Konfigurasi VLAN, access port, dan trunk port berhasil diterapkan untuk membangun segmentasi jaringan, sedangkan implementasi *Router-on-a-Stick* memungkinkan komunikasi antar VLAN serta distribusi alamat IP melalui DHCP. Konfigurasi alamat IP pada perangkat Cisco, Huawei, dan MikroTik juga berhasil membentuk konektivitas antar jaringan. Selain itu, penerapan routing dinamis OSPF pada lingkungan multivendor memungkinkan pertukaran informasi routing secara otomatis dan komunikasi *end-to-end* antar jaringan. Pengujian *failover* menunjukkan bahwa OSPF mampu melakukan perpindahan jalur secara otomatis saat terjadi gangguan dan kembali memilih jalur utama setelah koneksi dipulihkan. Melalui praktikum ini, praktikan memperoleh pemahaman mengenai konfigurasi jaringan multivendor, penerapan routing dinamis, serta pentingnya redundansi jaringan untuk meningkatkan keandalan dan ketersediaan layanan.

5 Lampiran



Gambar 58: Dokumentasi Praktikum